# Graphics Programming

# Scaling, Rotation, and Translation

GB. Tony Liang

# Scaling Transformations

- Multiples each component of a vector by a constant
  - $\mathrm{F}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} r \bullet x \\ s \bullet y \end{bmatrix} = \begin{bmatrix} r & 0 \\ 0 & s \end{bmatrix} \bullet \begin{bmatrix} x \\ y \end{bmatrix}$
  - F(): a scaling function that scales x-axis by *r*, and scales y-axis by *s*
    - E.g.,
      - Given a point at [10, 5], assume r=1.2, s=0.6, the new location of the point will be [12, 3]
      - $\mathrm{F}\left(\begin{bmatrix} 10 \\ 5 \end{bmatrix}\right) = \begin{bmatrix} 1.2 \bullet 10 \\ 0.6 \bullet 5 \end{bmatrix} = \begin{bmatrix} 1.2 & 0 \\ 0 & 0.6 \end{bmatrix} \bullet \begin{bmatrix} 10 \\ 5 \end{bmatrix} = \begin{bmatrix} 12 \\ 3 \end{bmatrix}$
- A scaling matrix only has values on the diagonal line

# Rotation Transformations

+ Rotates vectors by a constant angle ($\theta$) around the origin

+ 2D Rotation Matrix $\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$

  + E.g., Rotate vector <7,5> by 30 degree

  + $\begin{bmatrix} \cos(30) & -\sin(30) \\ \sin(30) & \cos(30) \end{bmatrix} \begin{bmatrix} 7 \\ 5 \end{bmatrix} = \begin{bmatrix} \sqrt{3/2} & -1/2 \\ 1/2 & \sqrt{3/2} \end{bmatrix} \begin{bmatrix} 7 \\ 5 \end{bmatrix} \approx \begin{bmatrix} 3.56 \\ 7.83 \end{bmatrix}$

# Rotations in 3D World

- Rotate around an axis, not just a point
- Rotating around origin in 2D = Rotating around z-axis in 3D
- Rotating around an axis does NOT move the axis
- Will only derive rotations around x-axis, y-axis, z-axis
  - In theory, we could rotate around any vector, but usually we only rotate object around the x-/y-/z-axis

# Rotation in 3D

+Around z-axis $\begin{bmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{bmatrix}$

+Around x-axis $\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{bmatrix}$

+Around y-axis $\begin{bmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & cos(\theta) \end{bmatrix}$

# 3D Rotation Matrix

$$
\begin{aligned}
R = R_z(\alpha)\, R_y(\beta)\, R_x(\gamma) &= \overset{\text{yaw}}{\begin{bmatrix} \cos\alpha & -\sin\alpha & 0 \\ \sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix}} \overset{\text{pitch}}{\begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}} \overset{\text{roll}}{\begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\gamma & -\sin\gamma \\ 0 & \sin\gamma & \cos\gamma \end{bmatrix}} \\
&= \begin{bmatrix} \cos\alpha\cos\beta & \cos\alpha\sin\beta\sin\gamma - \sin\alpha\cos\gamma & \cos\alpha\sin\beta\cos\gamma + \sin\alpha\sin\gamma \\ \sin\alpha\cos\beta & \sin\alpha\sin\beta\sin\gamma + \cos\alpha\cos\gamma & \sin\alpha\sin\beta\cos\gamma - \cos\alpha\sin\gamma \\ -\sin\beta & \cos\beta\sin\gamma & \cos\beta\cos\gamma \end{bmatrix}
\end{aligned}
$$

# Translation Transformations

- Adds constant values to each component of a vector
- $\mathrm{F}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x + m \\ y + n \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} m \\ n \end{bmatrix}$
- Translation is NOT a linear transformation!

# Translation is NOT a linear transformation

- Translation cannot be represented by a m-by-m matrix
- E.g., translating <x,y> by <2,0>
  - Translation matrix: $\begin{bmatrix} x+2 \\ y+0 \end{bmatrix}$
  - If the translation matrix is linear, we should be able to solve the following matrix
    - $\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+2 \\ y+0 \end{bmatrix}$
      - with a, b, c, d being the constant values.

# Translation is NOT a linear transformation

+Solve a, b, c, d values for

+$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x+2 \\ y+0 \end{bmatrix}$$

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax+by \\ cx+dy \end{bmatrix} = \begin{bmatrix} x+2 \\ y+0 \end{bmatrix}$$

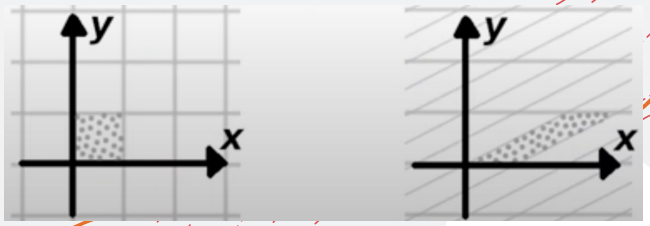
ax = x
by = 2
cx = 0
dy = y

a = 1
b = 2/y
c = 0
d = 1

What if we fixe b=2?

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} ax+by \\ cx+dy \end{bmatrix} = \begin{bmatrix} 1x+2y \\ 0x+1y \end{bmatrix} = \begin{bmatrix} x+2y \\ 0+y \end{bmatrix}$$

Shear Transformation



# Homogeneous Coordinates

- To present translation as a matrix transformation, embed the space being translated with a higher-dimensional space
- Set new coordinate = 1
  - E.g., (x, y, z) is identified with (x, y, z, 1)
    - recall: vec4 position = (x,y,z,1);
      - If we have (x, y, z, w), w≠1, OpenGL performs “perspective division)
        - (x, y, z, w) => (x/w, y/w, z/w)

# Homogeneous Coordinates for 2D Translation

$$\mathrm{F}\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} m \\ n \end{bmatrix} = \begin{bmatrix} x + m \\ y + n \end{bmatrix}$$

$$\mathrm{F}\left(\begin{bmatrix} x \\ y \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 & m \\ 0 & 1 & n \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} = \begin{bmatrix} x + m \\ y + n \\ 1 \end{bmatrix}$$

# Homogeneous Coordinates for 3D Translation

$$\begin{bmatrix} 1 & 0 & 0 & m \\ 0 & 1 & 0 & n \\ 0 & 0 & 1 & p \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix} = \begin{bmatrix} x + m \\ y + n \\ z + p \\ 1 \end{bmatrix}$$

# Translation, Rotation, and Scale

- When using homogeneous coordinates, translation, rotations, and scaling (affine transformations) can be represented by multiplying by a matrix of the form:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & m \\ a_{21} & a_{22} & a_{23} & n \\ a_{31} & a_{32} & a_{33} & p \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

**For rotation and scaling**

**For translation**

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & m \\ a_{21} & a_{22} & a_{23} & n \\ a_{31} & a_{32} & a_{33} & p \\ 0 & 0 & 0 & 1 \end{bmatrix}$$